M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 3 - Day 6 (Project) Report

Internship Program — Batch 2026

Submitted By:	Ghazi Muhammad Abdullah – AI26025
Department:	AI / ML
Submitted To:	Malik Mohazin Zahid
Date:	18 July 2026

1. Introduction

The Crop Recommendation System (v2) is an intelligent desktop application developed using Python, Scikit-learn, and PyQt5 to help farmers select the most suitable crop based on soil and environmental conditions. Selecting an appropriate crop is one of the most important factors affecting agricultural productivity. Many farmers rely on traditional knowledge and experience, which may not always produce the best results under changing climate and soil conditions. This project applies Machine Learning techniques to analyze agricultural parameters and recommend the most appropriate crop with high accuracy.

The application uses the K-Nearest Neighbors (KNN) classification algorithm to predict the best crop based on seven important input features: Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, Soil pH, and Rainfall. A user-friendly PyQt5 graphical interface allows users to enter agricultural data, validate inputs, and instantly receive crop recommendations along with prediction confidence, alternative crop suggestions, growing season, water requirements, and useful cultivation tips.

---

2. Objectives

- Develop an intelligent crop recommendation system using Machine Learning.
- Predict the most suitable crop based on soil nutrients and environmental conditions.
- Assist farmers in making scientifically informed crop selection decisions.
- Improve agricultural productivity by reducing unsuitable crop choices.
- Demonstrate the complete Machine Learning workflow, including data preprocessing, model training, evaluation, and prediction.
- Design a professional and easy-to-use desktop application using PyQt5.
- Provide reliable recommendations with high prediction accuracy and meaningful additional information.

---

3. Task Description

The project involved designing and implementing a complete Machine Learning-based desktop application for crop recommendation. A structured agricultural dataset containing information about different crops and their ideal growing conditions was prepared and used for model training.

The dataset was preprocessed by validating the data, scaling numerical features using StandardScaler, and encoding crop labels using LabelEncoder. The processed data was divided into training and testing sets before training a K-Nearest Neighbors (KNN) classifier. Cross-validation was used to determine the optimal value of K, and the trained model was evaluated to achieve approximately 98% prediction accuracy.

The trained model, scaler, and encoder were saved using Joblib and integrated into a PyQt5 desktop application. The GUI allows users to enter agricultural parameters, validate inputs, generate crop recommendations, display confidence scores, suggest alternative crops, and provide additional cultivation information in an easy-to-understand format.

---

4. Work Done

The project was completed by following the complete Machine Learning development process. Initially, a structured crop dataset covering 22 different crop types was prepared using realistic agricultural ranges for soil nutrients and environmental conditions. The dataset was organized with features including Nitrogen, Phosphorus, Potassium, Temperature, Humidity, Soil pH, Rainfall, and the target crop label.

Data preprocessing techniques such as feature scaling with StandardScaler and label encoding using LabelEncoder were applied before model training. The dataset was then divided into training and testing sets, and a weighted K-Nearest Neighbors (KNN) classifier was trained. Five-fold cross-validation was used to automatically determine the most suitable value of K, resulting in a highly accurate prediction model with approximately 98% test accuracy.

After training, the model, scaler, and encoder were saved and integrated into the desktop application. A modern PyQt5 graphical interface was developed with input validation to prevent incorrect entries and ensure stable operation. The application provides the recommended crop, prediction confidence, top three alternative crops, growing season, estimated water requirement, and useful growing tips. Additional features such as sample data loading, clearing inputs, and user-friendly warning messages were also implemented to improve usability and overall user experience.

---

5. Challenges Faced

Several challenges were encountered during the development of the project. Selecting the most suitable value of K for the KNN algorithm required experimentation and cross-validation to achieve the best prediction accuracy. Proper preprocessing of the dataset, including feature scaling and label encoding, was essential because KNN is sensitive to differences in feature values.

Designing a responsive and user-friendly PyQt5 interface while maintaining smooth interaction between the graphical interface and the Machine Learning model also required careful implementation. Another challenge was implementing comprehensive input validation to prevent invalid data from causing prediction errors. These challenges were successfully addressed through testing, debugging, and continuous refinement of both the Machine Learning model and the graphical application.

---

6. Key Learnings

This project provided valuable practical experience in developing a complete Machine Learning application from data preparation to deployment. I gained a deeper understanding of the K-Nearest Neighbors (KNN) classification algorithm, feature scaling, label encoding, model evaluation, and cross-validation techniques.

The project also strengthened my Python programming skills by applying object-oriented programming concepts and integrating Machine Learning models into a professional PyQt5 desktop application. I learned how to design user-friendly graphical interfaces, perform input validation, save and load trained models using Joblib, and build a reliable prediction system suitable for real-world agricultural applications. Overall, the project improved both my Machine Learning knowledge and software development skills.

7. Conclusion

The Crop Recommendation System (v2) successfully demonstrates how Machine Learning can support intelligent decision-making in agriculture. By analyzing soil nutrients and environmental conditions, the system accurately recommends the most suitable crop while providing additional cultivation information to assist users.

The project combines data preprocessing, model training, evaluation, and desktop application development into a complete AI solution. With an approximate prediction accuracy of 98%, the application provides reliable recommendations through a simple and intuitive interface. This project serves as an effective educational demonstration of applying Machine Learning to solve real-world agricultural problems and provides a strong foundation for future enhancements such as real-time weather integration, fertilizer recommendations, cloud connectivity, and support for larger agricultural datasets.

Intern Signature:	Ghazi	**Supervisor:**	
Name:	Ghazi Muhammad Abdullah	**Name:**	
Date:	18 July 2026	**Designation:**	